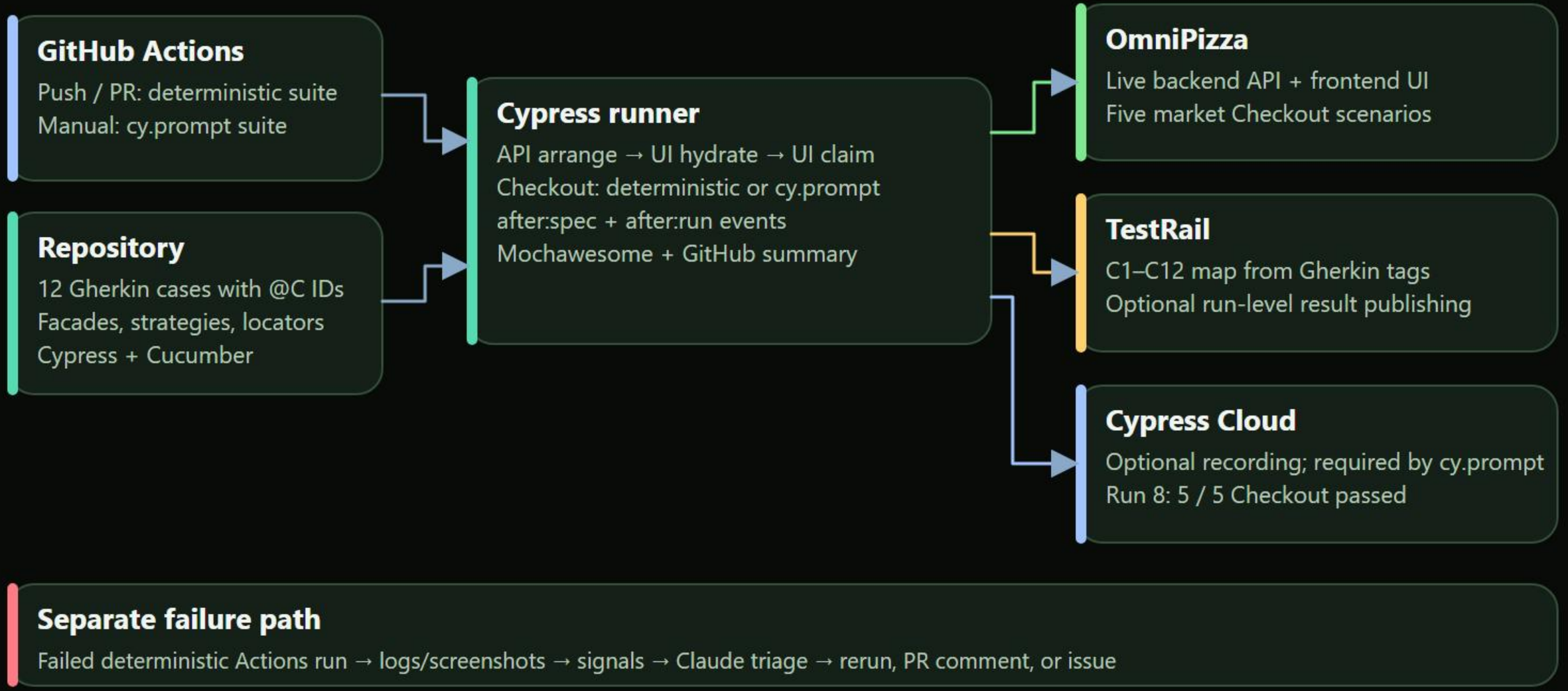
System architecture today

CURRENT

Versioned tests, live application, reporting, and an independent failure triage path



Separate failure path

Failed deterministic Actions run → logs/screenshots → signals → Claude triage → rerun, PR comment, or issue

One case, two contracts

CURRENT

The Gherkin tag identifies the TestRail case; the Cypress scenario remains the executable contract

TestRail case C9
Checkout / Mexico
Human-readable case
No pre-run read today

same C9

checkout.feature
@C9 on Mexico Examples row
Cucumber compiles this row
Scenario selects MX test data

AtomicScenario
arrangeViaApi → hydrateUi → assertUi
One API claim + one rendered UI claim

1 API arrange
Create and assert an API order
Keep its computed total as oracle

2 UI hydrate
Log in; complete a second order
Live frontend + backend interaction

3 UI claim
Compare confirmation total
against the API-derived total

After the run: TestRailObserver
after:spec collects results; after:run reads @C tags, maps passed/failed cases, calls add_run → add_results_for_cases → close_run.
Requires TESTRAIL_URL; pending/skipped tests are not submitted. Publishing errors are logged without changing the Cypress exit code.

Two UI adaptation paths

CURRENT

They solve different problems and have different evidence; do not present them as the same healer

Deterministic selector healing

Default CheckoutUiStrategy; no model call

What changes

MUTATE_DOM renames a data-testid
LocatorProxy tries the original selector
A11y fallback requires one unique match

Demonstrated

Five-market mutation demo and fallback pass
Selected fallback is logged in the run

Cypress cy.prompt

Opt-in CheckoutUiStrategy; Cloud-backed AI

How it runs

Natural-language steps in checkout.prompt.md
Country-specific target + runtime values
Two cy.prompt calls per scenario

Demonstrated

Five markets passed in recorded Cloud run 8
Healing of this specific DOM mutation is untested

CI communication map

CURRENT

Three distinct paths: automatic verification, manual AI Checkout, and failure-only agent triage

Push / PR
Typecheck
Auth + Catalog + Checkout
No Cloud recording by default

Deterministic suite
Mochawesome + screenshots
GitHub job summary
Optional TestRail, if configured

Success
Pipeline is green
Triage job is skipped

Failure evidence
Failed job logs + screenshots
Deterministic signal classifier
Claude JSON diagnosis

Current triage actions
Flake / cold start: capped rerun
Other: PR comment or GitHub Issue
No strict product-bug gate yet

Manual dispatch
cy-prompt-suite.yml
CYPRESS_RECORD_KEY secret

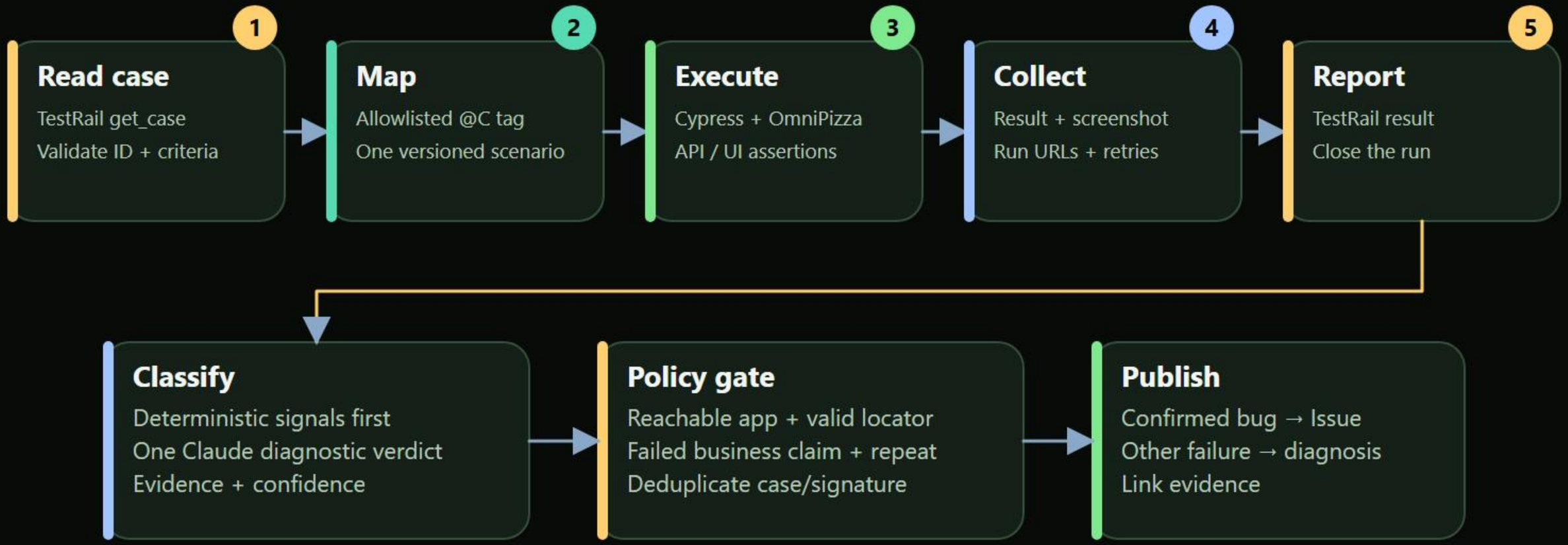
Checkout via cy.prompt
Five country examples
Recorded Chromium run

Cypress Cloud
Run results + Test Replay
CLI needs separate auth

Target agentic demo loop

PROPOSED

Planned connection for the talk; these boxes are not yet an implemented end-to-end pipeline



Already built: @C mapping, Cypress execution, TestRail result writer, Cloud recording, CI triage.

To build: case preflight, a single-case orchestration job, strict issue gate, and controlled product regression.